

PackLab: reproducible Percus–Yevick correlations for hard-sphere mixtures

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Abstract

PackLab is an open-source Python package centred on the analytical Percus–Yevick (PY) solution for multicomponent hard-sphere mixtures. It evaluates equilibrium-reference partial pair correlations and structure factors with unit-aware inputs and explicit Fourier-resolution diagnostics. A qualitative comparison with published real-space curves recovers the expected contact locations, curve ordering, and post-contact structure, while an external reciprocal-space implementation confirms mono-, bi-, and ternary-mixture results to within 9.6×10^{-10} . Compiled random sequential addition (RSA) tools complement the analytical workflow by constructing non-overlapping configurations, while an optional Metropolis Monte Carlo (MC) utility evolves valid configurations through fixed-volume trial moves.

Keywords: hard-sphere mixtures, Percus–Yevick, pair correlation, structure factor, random sequential addition, Monte Carlo

1. Motivation and significance

Hard-sphere-mixture correlation functions and structure factors are needed as reproducible structural inputs for colloids, particulate materials, porous media, disordered optical systems, and related scattering models [1, 2]. Reproducible studies require varying radius distributions, compositions, and

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Table 1: Code metadata.

Nr.	Code metadata description	Metadata
C1	Current code version	0.7.0
C2	Permanent link to code/repository used for this version	https://github.com/MartinPdeS/PackLab/tree/v0.7.0
C3	Permanent link to reproducible capsule	Not available; validation inputs and figure-generation scripts are versioned with the source.
C4	Legal code license	MIT License
C5	Code versioning system used	Git
C6	Software code languages, tools, and services used	Python, C++20, CMake, pybind11, NumPy, Matplotlib, and TypedUnit
C7	Compilation requirements, operating environments, and dependencies	Python 3.10 or later; CMake; a C++20 compiler; pybind11; OpenMP; NumPy, Matplotlib, and TypedUnit
C8	Link to developer documentation/manual	https://martinpdes.github.io/PackLab/docs/latest/index.html
C9	Support email for questions	martin.poinsinet.de.sivry@gmail.com

6 volume fractions while maintaining consistent physical units and Fourier-grid
7 resolution.
8 Dependent-scattering calculations for dense, polydisperse suspensions provide
9 a concrete example of this need. Attendu et al. combine single-particle
10 Mie theory with a multicomponent PY hard-sphere model to predict the
11 scattering properties of Intralipid across wavelength and concentration [3].
12 Such applications require the particle-size distribution, mixture correlation
13 model, and numerical settings to remain internally consistent.
14 PackLab provides this analytical reference in one documented Python inter-
15 face. Its core `analytical` namespace evaluates the PY hard-sphere-mixture
16 model [4, 5], retaining the mixture definition, dimensional inputs, and Fourier-
17 resolution choices with the result. The package is therefore suited to analyti-
18 cal parameter sweeps and structure-aware scattering models. Complemen-
19 tary RSA and MC workflows provide explicit configurations for downstream
20 coordinate-based calculations and carefully interpreted comparisons; they are
21 not substitutes for the equilibrium analytical reference.

22 1.1. State of the field

23 Existing software overlaps with individual parts of PackLab rather than with
24 its complete workflow. SasView is a broader small-angle-scattering analysis

Table 2: Executable software metadata.

Nr.	Software metadata description	Metadata
S1	Current software version	0.7.0
S2	Permanent link to executables of this version	https://pypi.org/project/PackLab/0.7.0/ ; https://anaconda.org/martinpdes/packlab
S3	Permanent link to reproducible capsule	Not available; the tagged source and versioned validation data reproduce the reported calculations.
S4	Legal software license	MIT License
S5	Computing platforms/operating systems	Linux, macOS, and Microsoft Windows
S6	Installation requirements and dependencies	<code>pip install PackLab</code> ; Python 3.10 or later; prebuilt wheels target CPython 3.11–3.13
S7	Link to user manual	https://martinpdes.github.io/PackLab/docs/latest/index.html
S8	Support email for questions	martin.poinsinet.de.sivry@gmail.com

environment that includes monodisperse and binary hard-sphere models using the PY closure [6, 7]. The mode-coupling package `mctpy` also contains an `hsmPY` component that evaluates reciprocal-space PY structure factors for mixtures specified by number densities and diameters [8]. This component overlaps with one PackLab output, but `mctpy`’s broader purpose and interface are different.

PackLab also supports scattering workflows through its optional PyMieSim integration, but its specific contribution is to keep multicomponent mixture construction, real-space partial pair correlations, reciprocal-space partial structure factors, physical units, and Fourier-grid diagnostics within one Python workflow. Its explicit-configuration tools additionally make it possible to generate RSA packings, evolve valid configurations with hard-sphere MC, and compare their estimated correlations with the analytical reference while preserving the distinction between the underlying models. PackLab is therefore intended for reproducible programmatic studies in which mixture definitions and numerical resolution must remain traceable across analytical, configuration, and optional scattering calculations; it is not a replacement for a general scattering-fitting or mode-coupling environment.

2. Software description

2.1. Percus–Yevick hard-sphere mixtures

The analytical workflow evaluates the multicomponent PY approximation for hard-sphere mixtures. `PercusYevickDomain` converts radii, number fractions, and total volume fraction into species densities. `PercusYevickSolver` then evaluates partial pair correlations $g_{ij}(r)$ and structure factors on a user-supplied or automatically constructed wavenumber grid.

PY is an approximate equilibrium reference. Its approximation error depends on mixture density, composition, and size ratio, while its numerical error depends separately on the inverse-transform grid. PackLab exposes both choices: automatic grids provide a documented default and the grid-construction interface permits explicit resolution studies. The digitised binary-mixture PY curves of Tsang et al. [9], used in Figure 3, provide an external visual comparison at matched radii and partial volume fractions.

2.2. Complementary simulation workflows

2.2.1. Random sequential addition

RSA proposes one sphere at a time. For a candidate centre $\mathbf{x}_i$ and radius a_i , PackLab accepts the candidate only if

$$\|\mathbf{x}_i - \mathbf{x}_j\| \geq a_i + a_j \quad (1)$$

for every previously accepted sphere j . Periodic simulations use the minimum-image convention. For a domain with volume V , the realised packing fraction is

$$\phi = \frac{1}{V} \sum_i \frac{4\pi a_i^3}{3}. \quad (2)$$

RSA is irreversible and history dependent: once a sphere is accepted, it remains fixed for the rest of the run, while rejected candidates are discarded and never reconsidered. Consequently, early accepted spheres can block later positions permanently, so the final configuration depends on the order of random proposals rather than being a sample from the equilibrium hard-sphere fluid [10]. PackLab makes this protocol explicit through independent stopping conditions, a stored random seed, periodic or finite domains, and constant, uniform, normal, log-normal, and discrete radius samplers. A spatial-neighbour grid limits overlap checks to nearby accepted spheres, preserving the RSA acceptance rule while avoiding an all-pairs test for every proposal.

2.2.2. Metropolis Monte Carlo

The optional MC utility takes a valid non-overlapping configuration, holds its particle count, radii, and class labels fixed, and proposes symmetric single-particle displacements. A proposal is accepted only when it remains within the domain and introduces no overlap, and the result reports displacement-move acceptance statistics. Users remain responsible for assessing burn-in, autocorrelation, and finite-size effects before interpreting saved states as equilibrium samples. Figure 4 provides an equilibrium-consistency check against the matched PY reference using independently initialized chains and an explicit burn-in stability diagnostic.

2.3. Software design and functionality

Performance-critical analytical and configuration kernels are implemented in C++20 and exposed as native Python extensions. The Python layer provides NumPy-compatible results, `TypedUnit`-based dimensional inputs, plotting helpers, type information, and executable examples. Native extensions are installed beside their public Python APIs, keeping imports such as `PackLab.monte_carlo` and `PackLab.analytical` stable and discoverable.

An analytical workflow first defines a mixture with `PercusYevickDomain`. The resulting domain is evaluated by `PercusYevickSolver`. Beginner-facing examples use `wavenumber="auto"`, which selects a zero-inclusive grid from the smallest particle radius and requested distance range. Users can instead supply a grid with `make_wavenumber_grid` when they need explicit control over radial resolution and Fourier sampling.

For explicit configurations, an RSA workflow combines a `PackingDomain`, a radius sampler, and `RSASOptions`. Calling `RSASimulator.run()` returns a `PackingResult`. The result contains centres, radii, class labels, packing statistics, and total or partial pair-correlation estimates. `MetropolisSimulator` accepts such a configuration with `MetropolisOptions` and reports displacement-move acceptance statistics. `PackingEstimator` reports ensemble correlation statistics together with insertion, rejection, runtime, and realised-packing diagnostics.

3. Illustrative examples

3.1. Complementary configuration workflows

Figure 2 shows a fixed-seed periodic RSA realisation of a binary mixture. The cross-section provides a direct geometric diagnostic of one generated

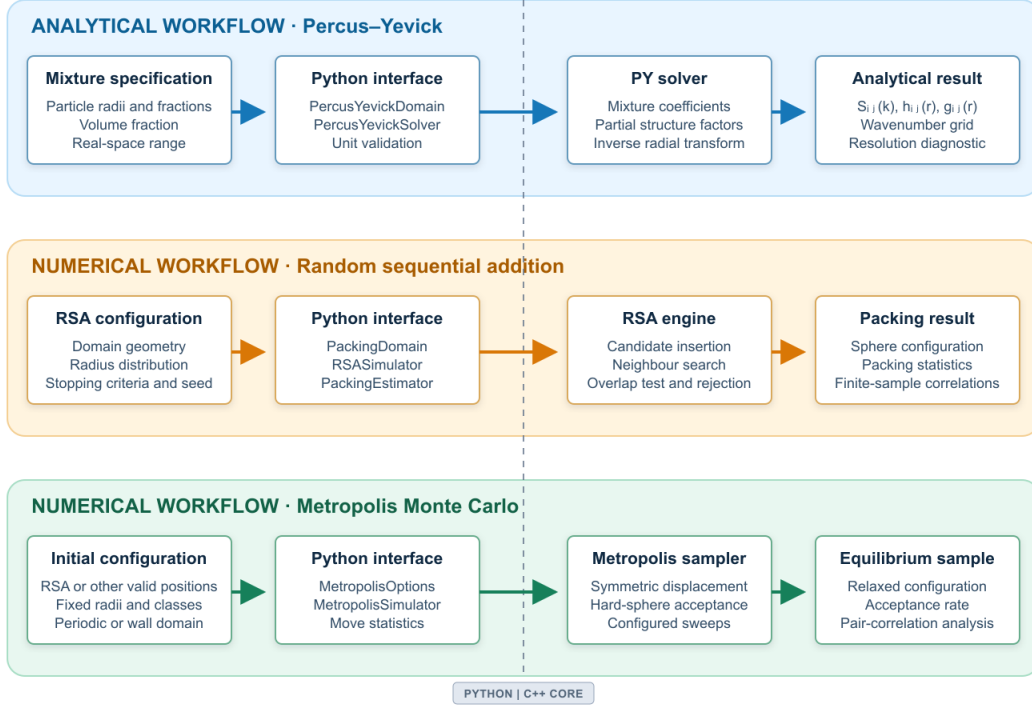


Figure 1: PackLab centres the PY equilibrium hard-sphere-mixture workflow and provides complementary numerical workflows. The analytical workflow evaluates pair correlations and structure factors for specified mixture inputs. The RSA and MC workflows create irreversible deposition configurations and evolve valid configurations through fixed-volume trial moves, respectively. All workflows have unit-aware Python APIs backed by native C++ kernels, but their results are compared and interpreted rather than treated as interchangeable.

109 configuration, while the correlation panel reports ensemble statistics for the
 110 same mixture. All quantitative figures are reproducible from the public API
 111 using the workflows distributed with the repository; the external reference
 112 data used for validation are versioned alongside the manuscript. Table 3
 113 collects the physical, stochastic, and numerical controls for all quantitative
 114 figures.

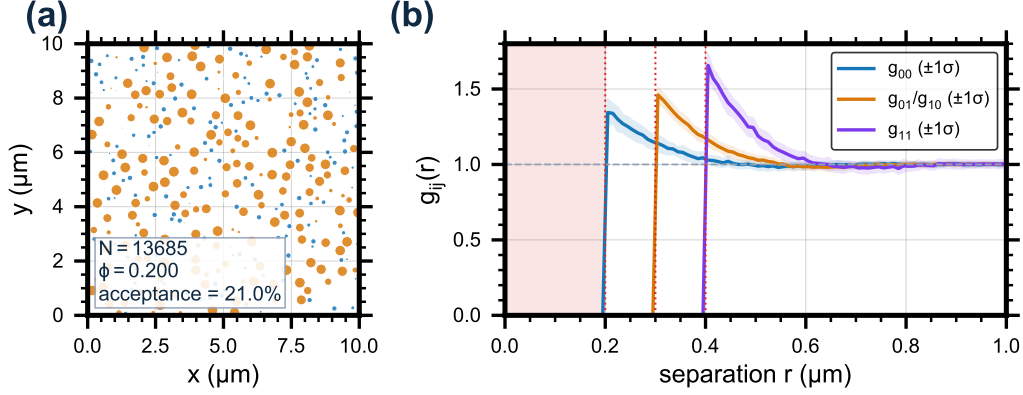


Figure 2: Reproducible periodic RSA demonstration for a two-radius mixture. The cubic domain has side length $10\ \mu\text{m}$; sphere radii are 100 and 200 nm with requested number fractions 0.65 and 0.35; the target packing fraction is 0.20; and the random seed is 20260828. (a) Central cross-section of the accepted three-dimensional configuration, with colour indicating radius class. (b) The unique partial pair correlations, $g_{00}(r)$, $g_{01}(r) = g_{10}(r)$, and $g_{11}(r)$, estimated from 64 independent RSA realisations. The legend identifies each ensemble-mean coloured line; its matching same-colour band shows $\pm 1\sigma$, where σ is the sample standard deviation across realisations. The red shaded interval is the exclusion range shared by all pairs, while the red dotted lines mark the three contact separations, 0.20, 0.30, and 0.40 μm . The dashed horizontal line marks the uncorrelated limit $g_{ij}(r) = 1$.

115 3.2. Percus–Yevick validation and numerical resolution

116 3.2.1. Analytical correlation and structure-factor conventions

117 The RSA estimator is used only for the complementary comparison below.
 118 It defines $g_{ij}(r_b)$ as the ratio between the observed ordered i – j pair count in
 119 radial shell b and the ideal-gas expectation for the same configuration. For
 120 species counts N_i and N_j , domain volume V , and shell volume ΔV_b , the latter
 121 is

$$E_{ij,b} = N_i (N_j - \delta_{ij}) \frac{\Delta V_b}{V}, \quad g_{ij}(r_b) = \frac{C_{ij,b}}{E_{ij,b}}, \quad (3)$$

122 where δ_{ij} is the Kronecker delta and $C_{ij,b}$ is the observed ordered count. The
 123 radial bins are spherical shells with centres $r_b = (b + 1/2)\Delta r$ and extend
 124 to half the shortest box dimension; periodic estimates use minimum-image
 125 separations. When pair sampling is requested, the observed histogram is
 126 rescaled by the ratio of all unordered pairs to examined unordered pairs before
 127 this normalization.

128 For the analytical workflow, $h_{ij}(r) = g_{ij}(r) - 1$ and $H_{ij}(k) = \sqrt{\rho_i \rho_j} \tilde{h}_{ij}(k)$ is

129 the density-scaled reciprocal-space total correlation. We use the corresponding
 130 density-scaled partial structure-factor convention

$$S_{ij}(k) = \delta_{ij} + H_{ij}(k). \quad (4)$$

131 This convention makes both the pair-correlation and structure-factor com-
 132 parisons dimensionless and identifies the density factors required when trans-
 133 forming between them.

134 The PY calculation uses an inverse radial Fourier transform, so its real-space
 135 result depends on wavenumber-grid extent and sampling density. For an
 136 isotropic mixture, the real-space total correlation is recovered from $H_{ij}(k)$
 137 through

$$h_{ij}(r) = \frac{1}{2\pi^2 \sqrt{\rho_i \rho_j}} \int_0^\infty k^2 H_{ij}(k) \text{sinc}(kr) dk. \quad (5)$$

138 At the maximum requested distance r_{max} , the $\text{sinc}(kr)$ factor has wavenumber
 139 period $2\pi/r_{\text{max}}$, which is the most rapidly varying kernel over the requested
 140 distance range. A truncated grid omits high-wavenumber content, whereas
 141 a coarse grid under-samples oscillations in the sinc kernel. Either case can
 142 produce a visually smooth but numerically misleading correlation function.
 143 For `wavenumber="auto"`, PackLab uses a real-space resolution of one twenti-
 144 eth of the smallest radius and twelve samples per sinc-kernel oscillation at
 145 the maximum requested distance. When a user-provided grid has fewer than
 146 eight samples per oscillation, PackLab raises a runtime warning. These are
 147 numerical defaults, not physical constants, and the grid-construction API
 148 allows stricter choices.

149 *3.2.2. Independent analytical comparisons*

150 Figure 3 compares PackLab with independent real- and reciprocal-space PY
 151 results. Panels (a) and (b) compare PackLab’s mono- and binary-mixture
 152 solutions with manually digitised PY curves from Tsang et al. [9] at matched
 153 volume fractions and radius ratio. The expected hard-core contact locations,
 154 decay toward the uncorrelated limit, and damped post-contact structure
 155 agree visually across both monodisperse curves and all three binary partial
 156 correlations. Because the reference coordinates were manually recovered from
 157 printed plots, their pointwise uncertainty is not sufficiently controlled for a
 158 meaningful numerical error metric. The comparison is therefore interpreted
 159 qualitatively and is used to check the principal physical features of the
 160 published solutions rather than to estimate implementation error.

161 Panel (c) provides a higher-precision reciprocal-space check against versioned
 162 reference data generated with the independently implemented **hsmPY** compo-
 163 nent of **mctpy** version 0.0.2.post1 [8]. Both calculations use the density-scaled
 164 convention of Equation 4, and the checked-in reference contains 601 wavenum-
 165 bers from 0 to $30 \mu\text{m}^{-1}$. For the displayed binary mixture, the maximum
 166 absolute difference across all partial structure factors is 9.58×10^{-10} . The
 167 same reproducible comparison gives maximum differences of 2.32×10^{-11} and
 168 5.97×10^{-10} for the monodisperse and ternary cases, respectively.

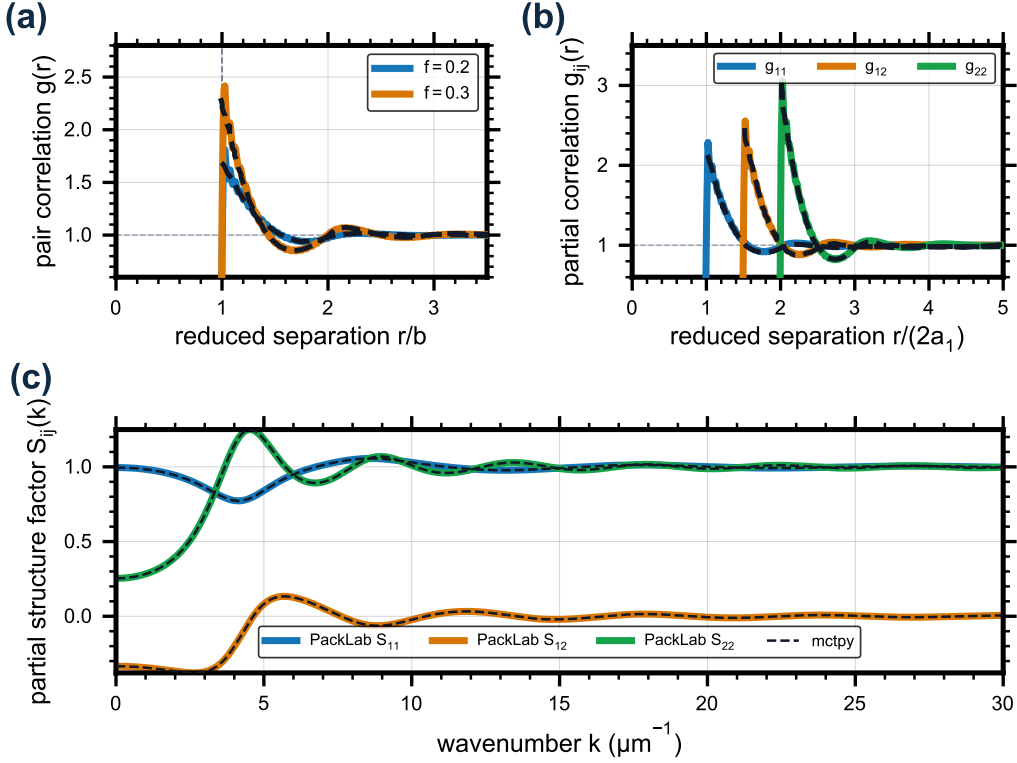


Figure 3: Independent validation of PackLab's real- and reciprocal-space PY outputs. (a) Monodisperse PackLab correlations (coloured solid lines) and digitised Tsang et al. references [9] (black dashed lines) at volume fractions $f = 0.2$ and $f = 0.3$, plotted against r/b . (b) Binary-mixture PackLab correlations (coloured solid lines) and digitised references (black dashed lines) for $a_2 = 2a_1$ and partial volume fractions $f_1 = 0.2$, $f_2 = 0.1$, plotted against $r/(2a_1)$. (c) PackLab (coloured solid curves) and $mctpy$ (black dashed curves) partial structure factors for radii 0.35 and 0.70 μm , number fractions 0.65 and 0.35, and total volume fraction 0.25.

169 *3.2.3. Complementary RSA comparison and numerical convergence*

170 Figure 4 distinguishes model comparison from numerical verification. Panel (a)
171 compares all unique partial correlations from a 100-realisation RSA ensemble
172 with PY evaluated for the same binary-mixture inputs and the same radial
173 coordinates. The differences are expected physical consequences of comparing
174 finite, irreversible RSA with an equilibrium PY reference, rather than a
175 numerical-error measure. Panel (b) compares PY with 12 independently
176 initialized MC chains after 600 burn-in sweeps. The mean move-acceptance
177 rate is 0.782, and the post-contact root-mean-square (RMS) differences are
178 0.045, 0.027, and 0.032 for g_{00} , g_{01} , and g_{11} , respectively. Between 300 and
179 600 sweeps, 94.3% of resolved correlation values agree within the combined
180 95% uncertainty of the two ensemble means. Because PY is approximate, this
181 is an equilibrium-consistency check rather than an exact-error measurement.
182 Panel (c) isolates numerical convergence by measuring the PY result against
183 a separately specified, much finer wavenumber grid.

184 **4. Impact**

185 PackLab turns a collection of mixture definitions, analytical expressions,
186 Fourier transforms, configuration generators, and statistical estimators into a
187 reusable and inspectable research workflow. Its principal impact is method-
188 ological reproducibility: a study can carry the same radii, composition, volume
189 fraction, units, and reciprocal-space resolution from mixture construction
190 to pair correlations, structure factors, and optional dependent-scattering
191 calculations. The Intralipid model of Attendu et al. [3] illustrates the class of
192 research enabled by this workflow, in which measured particle-size informa-
193 tion is combined with multicomponent hard-sphere correlations to evaluate
194 concentration-dependent optical scattering. The public Python interface and
195 executable examples also make systematic parameter sweeps and indepen-
196 dent numerical checks accessible without requiring users to reimplement the
197 Lebowitz mixture solution or its transforms.

198 *4.1. Quality control and availability*

199 PackLab tests analytical hard-core exclusion, automatic-grid selection, grid
200 warnings, and convergence through its public API. Separate tests cover sam-
201 plers, packing validity, periodic boundaries, and Metropolis move validation.
202 Executable documentation examples are organised around analytical, RSA,

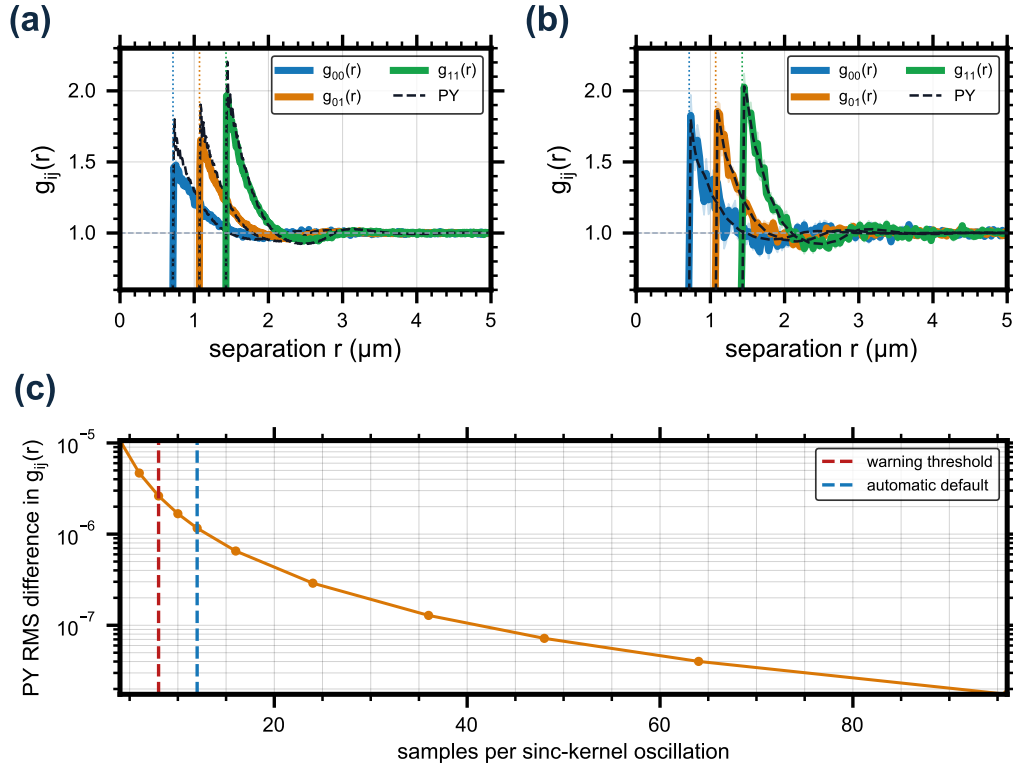


Figure 4: Matched configuration–PY comparisons and PY grid convergence for a binary mixture with radii 0.357 and 0.714 μm , equal number fractions, and total volume fraction 0.25. (a) RSA ensemble means (coloured solid curves) with one-standard-error bands and PY references (black dashed curves) for the three unique partial correlations; dotted lines mark hard-core contact distances. (b) Means from 12 independently initialized MC chains after 600 burn-in sweeps, with one-standard-error bands and matched PY references. (c) RMS difference from a 512-samples-per-oscillation PY reference as the wavenumber grid is refined; red and blue dashed lines mark the warning threshold (8) and automatic default (12), respectively.

Table 3: Reproducibility controls for the quantitative figures. Figure numbers refer to the rendered manuscript.

Figure	Mixture and geometry	Stochastic controls	Numerical and reference controls
2	Radii 100 and 200 nm; number fractions 0.65 and 0.35; periodic cube of side 10 μm ; target $\phi = 0.20$.	Seed 20260828; 64 independent realisations; 500 radial bins.	Pair correlations to 2 μm ; maximum attempts and consecutive rejections of 100000 and 20000.
3	Tsang mono- and binary-mixture inputs; <code>mctpy</code> mono-, bi-, and ternary-mixture checks.	Deterministic analytical calculations.	Automatic grids for $g(r)$; digitised curves assessed qualitatively; 601 points over $0 \leq k \leq 30 \mu\text{m}^{-1}$ for the quantitative $S_{ij}(k)$ check.
4	Radii 0.357 and 0.714 μm ; equal number fractions; $\phi = 0.25$; periodic cubes of side 30 μm (RSA) and 15 μm (MC).	RSA seed 2026, 100 realisations, and 1500 bins; 12 MC chains with seeds 4100–4111 and 5100–5111, 600 burn-in sweeps, and 250 bins.	MC displacement 0.12 μm ; grid resolutions 4–96 samples per oscillation; 512-samples-per-oscillation reference; $r_{\text{max}} = 10 \mu\text{m}$ and 1000 radial points.

MC, and comparison workflows. Continuous integration (CI) builds the native extensions and runs the test suite for CPython 3.11–3.13. PackLab version 0.7.0 requires Python 3.10 or later, is released under the MIT License, and is available from Python Package Index (PyPI) and conda. Source code, documentation, issue tracking, and release history are available at <https://github.com/MartinPdeS/PackLab>. Release-specific citation metadata is maintained in `.zenodo.json`. Archived releases are available from the Zenodo record at [10.5281/zenodo.20207810](https://zenodo.org/record/20207810).

4.2. Limitations

The current scope is three-dimensional, spherical, hard-core particles in cuboidal domains. RSA should not be interpreted as an equilibrium-fluid simulator; MC requires explicit equilibration assessment; and PY should not be interpreted as a measured structure or a general interaction model. The practical limits of both workflows depend on the requested mixture and numerical settings. For RSA, insertion becomes progressively less efficient as the requested packing fraction increases; finite domains, broad radius

219 distributions, and stringent stopping conditions can prevent the requested
 220 target from being reached. Consequently, users should report the realised
 221 packing fraction, stopping diagnostics, random seed, ensemble size, and
 222 domain size rather than treating a requested target as an achieved state.
 223 Pair-correlation estimates from finite configurations additionally depend on
 224 binning and are subject to finite-size uncertainty, particularly near the largest
 225 separations accessible in a domain. For MC, users should assess burn-in,
 226 autocorrelation, proposal acceptance, and finite-size effects before treating
 227 separated chain states as equilibrium samples.
 228 For PY, the approximation error depends on density, composition, and
 229 particle-size ratio, and is separate from Fourier-grid discretisation error. The
 230 automatic grid and under-resolution warning are useful safeguards, but they
 231 do not establish convergence for every mixture, especially when near-contact
 232 structure is sharp, size ratios are extreme, or correlations are required over a
 233 long distance range. Users should refine the wavenumber grid and compare
 234 against an independently chosen equilibrium benchmark when the analytical
 235 result informs a quantitative conclusion. PackLab’s separate namespaces,
 236 explicit input definitions, recorded stochastic settings, and grid diagnostics
 237 are designed to make these limitations visible at the point of use.

238 **5. Conclusions**

239 PackLab provides a unit-aware Python interface to multicomponent PY pair
 240 correlations and partial structure factors, supported by native C++20 kernels
 241 and explicit Fourier-grid diagnostics. Qualitative real-space comparisons
 242 recover the principal features of published hard-sphere-mixture curves, while
 243 reciprocal-space results agree with a separately implemented reference to
 244 within 9.6×10^{-10} for the tested mono-, bi-, and ternary mixtures. The
 245 package’s RSA generator and MC sampler extend the workflow to explicit
 246 configurations without treating their outputs and the analytical reference as
 247 interchangeable. Together, these facilities support reproducible structural
 248 inputs for hard-particle and dependent-scattering studies while retaining the
 249 numerical and physical assumptions needed to interpret each result.

250 **CRedit authorship contribution statement**

251 Martin Poinsinet de Sivry-Houle: Conceptualization, Methodology, Software,
 252 Validation, Visualization, Writing—original draft, Writing—review and editing.

253 Xavier Attendu: Methodology, Validation, Writing–review and editing. Ton
254 G. van Leeuwen: Supervision, Writing–review and editing. Edwin van der
255 Pol: Supervision, Writing–review and editing.

256 **Funding**

257 This research received no specific grant from funding agencies in the public,
258 commercial, or not-for-profit sectors.

259 **Declaration of competing interest**

260 The authors declare that they have no known competing financial interests
261 or personal relationships that could have appeared to influence the work
262 reported in this article.

263 **Data availability**

264 All source code, tests, figure-generation scripts, and numerical reference data
265 used in this article are available in the PackLab repository and its versioned
266 release archive.

267 **Declaration of generative AI and AI-assisted technologies in the** 268 **manuscript preparation process**

269 During the preparation of this work, the authors used OpenAI Codex to assist
270 with manuscript organization, language editing, and software-maintenance
271 tasks. The authors reviewed and edited the output and take full responsibility
272 for the content of the publication.

273 **Acknowledgements**

274 The authors thank the open-source scientific Python ecosystem and the users
275 who provide feedback, issues, and reproducible examples.

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